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crystALL: Purely data-driven automated structure prediction for unidentified chemical compounds

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Summary

- (2 sentences) Motivation for materials discovery
  - CALPHAD (Olson & Liu, 2023)
  - Going beyond stability. It is not enough (Zunger, 2019) because of the complexities involved in the synthesis of new materials. Phase coexistance can take properties to extreme, e.g. infinitely negative thermal expansion (Liu et al., 2022).
- Typically done through combinatoric search based on protoypes and DFT
  - Proposing new structures for compounds is not a new idea based on geometrical constraints (Bail, 2005)
  - AFLOW Database (Toher et al., 2018) built by expert-intuition driven population of AFLOW Prototype Library (Mehl et al., 2016) with new chemistries in a systematic way
  - Similarity driven prediction of substitutions (Wang et al., 2021)
  - Compositional complexity grows quickly with dimensionality, considering thhat substitutions (A. M. Krajewski et al., 2024)
- (3 sentences) Past deployment on the Nd-Bi system (Im et al., 2022) and Figure 7a of (Im et al., 2022) showing excellent ordering of the candidates when evaluated vs different DFT methods. Past deployment in Al-Fe (Shang et al., 2021).

Statement of need

(statement of need)

- Motivation slightly reiterating the summary with more focus on impact and need
- (**Rushi**, 3 sentences) (Importance of Ru-Zn in the study of catalysts. Importance of KCl-LaCl3 system in the study of molten salts. How `crystALL` is continuously used in the lab to predict structures of unknown compounds and if the automated experience has been positive.) Ru-based catalysts are recognized as highly effective benzene partial hydrogenation catalysts (Zhang et al., 2013). To enhance hydrogenation selectivity, the strategy involves designing diverse active site ensembles in intermetallic catalysts (Dasgupta et al., 2022). Using crystALL, over 120, 000 configurations of Ru-Zn intermetallic compounds have been predicted. We are currently examining their structures and stability through pySIPFENN (A. M. Krajewski et al., 2022), alongside evaluating active site ensembles, aimed to design novel catalysts with high activity and selectivity. Additionally, crystALL serves as a valuable tool for identifying structures of complex molten salts, particularly where experimental data are limited or conflicting. For example, crystALL facilitated the prediction of structures for compounds KLa3Cl10 and K3La5Cl18 in the KCl-LaCl3 system, thereby validating their stability reported in various

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literature sources (Seifert et al., 1985) (Song & Zheng, 1995).
- (**Adam**, 1 sentence) Conclusion on the importance of automation for user tool.

Methods

MPDD Database

Contains: - OQMD (Shen et al., 2022) - AFLOW Database (Toher et al., 2018) - Materials Project (Jain et al., 2013) - NIST-JARVIS (Choudhary et al., 2020) - COD (Gražulis et al., 2019) - GNoME (Merchant et al., 2023)

API: - native pymongo - OPTIMADE (Evans et al., 2024)

- use of `pymatgen` (Ong et al., 2013) for data processing

Predictive Modeling Pipeline

(about the pySIPFENN, MPDD-ALIGNN, and other models)

- pySIPFENN software (A. M. Krajewski et al., 2022) and models on Zenodo (A. Krajewski et al., 2020)
- ALIGNN (Choudhary & DeCost, 2021)

(schematic of crystALL software)

Compound-Based Structure Prediction

(about the structure prediction)

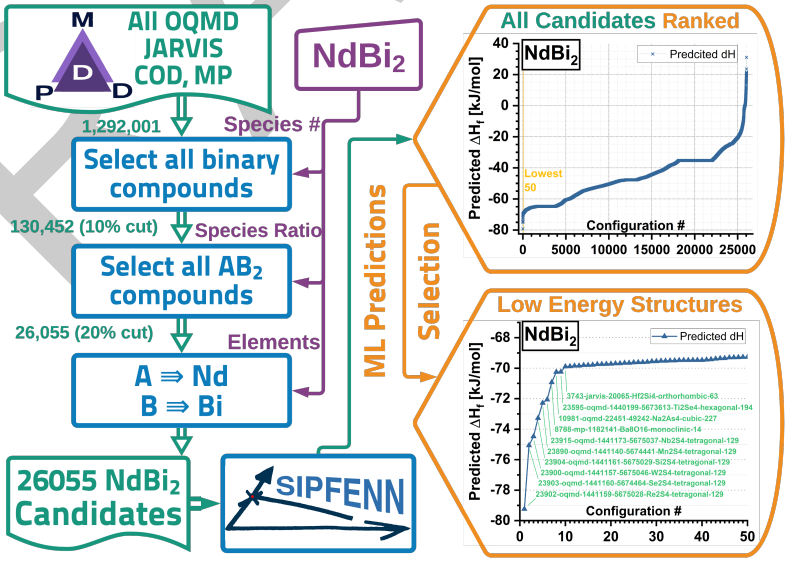


Figure 1: Compound-Based Structure Search. Reproduced from (Im et al., 2022) under permission.

Composition-Based Structure Prediction

(uncertainty in compound composition extends the previous method)

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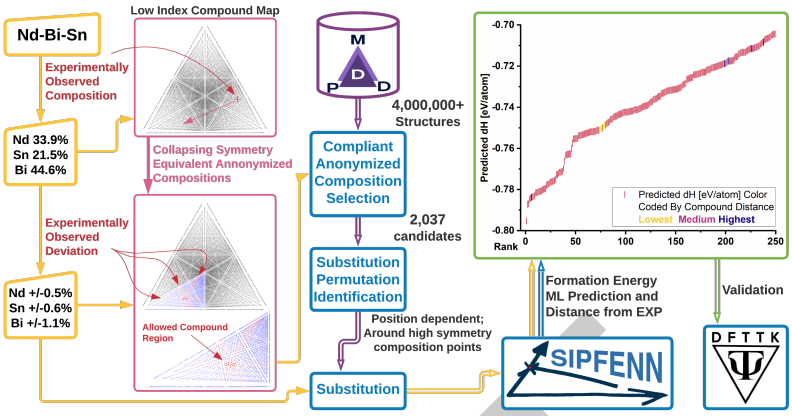


Figure 2: Composition-Based Structure Search.

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- All Phases Research Lab members who provided feedback on the software

Bail, A. L. (2005). Inorganic structure prediction with GRINSP. *Journal of Applied Crystallography*, *38*, 389–393. https://doi.org/10.1107/S0021889805002384

Choudhary, K., & DeCost, B. (2021). Atomistic line graph neural network for improved materials property predictions. *Npj Computational Materials*, *7*, 185. https://doi.org/10.1038/s41524-021-00650-1

Choudhary, K., Garrity, K. F., Reid, A. C. E., DeCost, B., Biacchi, A. J., Walker, A. R. H., Trautt, Z., Hattrick-Simpers, J., Kusne, A. G., Centrone, A., Davydov, A., Jiang, J., Pachter, R., Cheon, G., Reed, E., Agrawal, A., Qian, X., Sharma, V., Zhuang, H., … Tavazza, F. (2020). The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design. *Npj Computational Materials*, *6*, 1–13. https://doi.org/10.1038/s41524-020-00440-1

Dasgupta, A., He, H., Gong, R., Shang, S.-L., Zimmerer, E. K., Meyer, R. J., Liu, Z.-K., Janik, M. J., & Rioux, R. M. (2022). Atomic control of active-site ensembles in ordered alloys to enhance hydrogenation selectivity. *Nature Chemistry*, 1–7. https://doi.org/10.1038/s41557-021-00855-3

Evans, M. L., Bergsma, J., Merkys, A., Andersen, C. W., Andersson, O. B., Beltrán, D., Blokhin, E., Boland, T. M., Balderas, R. C., Choudhary, K., Díaz, A. D., García, R. D., Eckert, H., Eimre, K., Montero, M. E. F., Krajewski, A. M., Mortensen, J. J., Duarte, J. M. N., Pietryga, J., … Armiento, R. (2024). *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*. https://doi.org/10.48550/arXiv.2402.00572

Gražulis, S., Merkys, A., Vaitkus, A., Chateigner, D., Lutterotti, L., Moeck, P., Quiros, M., Downs, R. T., Kaminsky, W., & Bail, A. L. (2019). Crystallography open database: History, development, and perspectives. In *Materials Informatics* (pp. 1–39). Wiley. https://doi.org/10.1002/9783527802265.ch1

Im, S., Shang, S. L., Smith, N. D., Krajewski, A. M., Lichtenstein, T., Sun, H., Bocklund, B. J., Liu, Z. K., & Kim, H. (2022). Thermodynamic properties of the nd-bi system via emf measurements, DFT calculations, machine learning, and CALPHAD modeling. *Acta Materialia*, *223*, 117448. https://doi.org/10.1016/J.ACTAMAT.2021.117448

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Jain, A., Ong, S. P., Hautier, G., Chen, W., Richards, W. D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., & Persson, K. A. (2013). Commentary: The materials project: A materials genome approach to accelerating materials innovation. In *APL Materials* (Vol. 1, p. 011002). American Institute of Physics Inc. https://doi.org/10.1063/1.4812323

Krajewski, A. M., Beese, A. M., Reinhart, W. F., & Liu, Z.-K. (2024). *Efficient generation of grids and traversal graphs in compositional spaces towards exploration and path planning exemplified in materials*. https://doi.org/10.48550/arXiv.2402.03528

Krajewski, A. M., Siegel, J. W., Xu, J., & Liu, Z. K. (2022). Extensible structure-informed prediction of formation energy with improved accuracy and usability employing neural networks. *Computational Materials Science*, *208*. https://doi.org/10.1016/J.COMMATSCI.2022.111254

Krajewski, A., Siegel, J., Xu, J., & Liu, Z.-K. (2020). *Neural networks for structure-informed prediction of formation energy (employed in SIPFENN)*. Zenodo. https://doi.org/10.5281/zenodo.4006802

Liu, Z.-K., Wang, Y., & Shang, S.-L. (2022). Zentropy theory for positive and negative thermal expansion. *Journal of Phase Equilibria and Diffusion*. https://doi.org/10.1007/s11669-022-00942-z

Mehl, M. J., Hicks, D., Toher, C., Levy, O., Hanson, R. M., Hart, G., & Curtarolo, S. (2016). *The AFLOW library of crystallographic prototypes*. http://arxiv.org/abs/1607.02532

Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *80 | Nature |*, *624*. https://doi.org/10.1038/s41586-023-06735-9

Olson, G. B., & Liu, Z. K. (2023). Genomic materials design: CALculation of PHAse dynamics. *Calphad*, *82*, 102590. https://doi.org/10.1016/j.calphad.2023.102590

Ong, S. P., Richards, W. D., Jain, A., Hautier, G., Kocher, M., Cholia, S., Gunter, D., Chevrier, V. L., Persson, K. A., & Ceder, G. (2013). Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, *68*, 314–319. https://doi.org/10.1016/j.commatsci.2012.10.028

Seifert, H. J., Fink, H., & Thiel, G. (1985). Thermodynamic properties of double chlorides in the systems ACl/LaCl3 (A Na, K, Rb, Cs). *Journal of The Less-Common Metals*, *110*(1-2), 139–147. https://doi.org/10.1016/0022-5088(85)90315-7

Shang, S.-L., Sun, H., Pan, B., Wang, Y., Krajewski, A. M., Banu, M., Li, J., & Liu, Z.-K. (2021). Forming mechanism of equilibrium and non-equilibrium metallurgical phases in dissimilar aluminum/steel (al–fe) joints. *Scientific Reports*, *11*, 24251. https://doi.org/10.1038/s41598-021-03578-0

Shen, J., Griesemer, S. D., Gopakumar, A., Baldassarri, B., Saal, J. E., Aykol, M., Hegde, V. I., & Wolverton, C. (2022). Reflections on one million compounds in the open quantum materials database (OQMD). *Journal of Physics: Materials*, *5*, 031001. https://doi.org/10.1088/2515-7639/ac7ba9

Song, X., & Zheng, C. (1995). An Investigation on the Phase Diagram of Ternary System LaCl3-KCl-LiCl in Chinese). *Acta Chimica Sinica*, *53*, 978–984.

Toher, C., Oses, C., Hicks, D., Gossett, E., Rose, F., Nath, P., Usanmaz, D., Ford, D. C., Perim, E., Calderon, C. E., Plata, J. J., Lederer, Y., Jahnátek, M., Setyawan, W., Wang, S., Xue, J., Rasch, K., Chepulskii, R. V., Taylor, R. H., … Curtarolo, S. (2018). The AFLOW fleet for materials discovery. In *Handbook of Materials Modeling* (pp. 1–28). Springer International Publishing. https://doi.org/10.1007/978-3-319-42913-7_63-1

Wang, H.-C., Botti, S., & Marques, M. A. L. (2021). Predicting stable crystalline compounds using chemical similarity. *Npj Computational Materials*, *7*, 12. https://doi.org/10.1038/

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s41524-020-00481-6

Zhang, P., Wu, T., Jiang, T., Wang, W., Liu, H., Fan, H., Zhang, Z., & Han, B. (2013). Ru–Zn supported on hydroxyapatite as an effective catalyst for partial hydrogenation of benzene. *Green Chemistry*, *15*(1), 152–159. https://doi.org/10.1039/C2GC36596K

Zunger, A. (2019). Beware of plausible predictions of fantasy materials. *Nature 2021 566:7745*, *566*, 447–449. https://doi.org/10.1038/d41586-019-00676-y

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